B3.3 Maintaining internal environments

Summary

Homeostasis is crucial to the regulation of internal environments and enables organisms to adapt to change, both internally and externally. Internal temperature, blood sugar levels and osmotic balance are regulated by a number of organs and systems working together.

Underlying knowledge and understanding

Learners will build on the knowledge and understanding gained in section 3.1 about coordination and control when considering the topics in this section.

Common misconceptions

Learners often confuse type 1 and type 2 diabetes, and the effective treatments for each. The effect of ADH on the permeability of the kidney tubules is often confused.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers. All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
BM3.3i	extract and interpret data from graphs, charts and tables	M2c

Topic content			Opportunities to cover:		Practical suggestions
Learning outcomes		To include	Maths	Working scientifically	
B3.3a	explain the importance of maintaining a constant internal environment in response to internal and external change	allowing metabolic reactions to proceed at appropriate rates		WS1.4a	Research into hypothermia.
B3.3b ☑	describe the function of the skin in the control of body temperature	detection of external temperature, sweating, shivering, change to blood flow in terms of vasoconstriction and vasodilation		WS2a, WS2b, WS2c, WS2d	Demonstration of the cooling effect of sweating using alcohol based surgical wipes. (PAG B6) Investigation into heat loss by using microwaved plasticine shapes/model 'animals' by using a UV heat camera/thermometers. (PAG B6)
B3.3c	explain how insulin controls blood sugar levels in the body		M2g		

B3.3d	**explain how glucagon interacts with insulin to control blood sugar levels in the body**		M2c	WS2a, WS2b, WS2c, WS2d	Investigations into the glucose content of artificial urine to diagnose diabetes, using e.g. Clinistix. (PAG B6)
B3.3e	compare type 1 and type 2 diabetes and explain how they can be treated				
B3.3f ☑	explain the effect on cells of osmotic changes in body fluids	higher, lower or equal water potentials leading to lysis or shrinking (no mathematical use of water potentials required)		WS2a, WS2b, WS2c, WS2d	Demonstration of the different water potentials on different cells. (PAG B6, PAG B8)
B3.3g ☑	describe the function of the kidneys in maintaining the water balance of the body	varying the amount and concentration of urine and hence water excreted		WS1.3b, WS2a, WS2b, WS2c, WS2d	Investigation of the structure of the structure of a kidney by dissection and the application of H_2O_2 to visualise the nephrons. (PAG B6, PAG B8) Investigations into the glucose content of artificial urine to diagnose diabetes, using e.g. Clinistix. (PAG B6)
B3.3h ☑	describe the gross structure of the kidney and the structure of the kidney tubule	Bowman's capsule, proximal convoluted tubule, loop of Henlé and collecting duct			
B3.3i ☑	**describe the effect of ADH on the permeability of the kidney tubules**	**amount of water reabsorbed and negative feedback**		WS2a, WS2b, WS2c, WS2d	Investigation of the different sections of a nephron and the composition of the filtrate from each area. (PAG B2, PAG B6, PAG B8)
B3.3j ☑	**explain the response of the body to different temperature and osmotic challenges**	**challenges to include high sweating and dehydration, excess water intake, high salt intake responses to include mechanism of kidney function, thirst**			Research into sports drinks and evaluation into which is best for athletes. (PAG B2, PAG B6, PAG B8)